

ANDY LYEK

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EDUCATION

Stanford University

Bachelor of Science in Engineering, Electrical Engineering & Computer Science

Relevant Coursework: CS107: Computer Systems, CS109: Probability for CS; EE108: Digital System Design

Stanford, CA

Expected June 2028

EXPERIENCE

Software Engineer Intern

Northrop Grumman

June 2026 – Aug 2026 (Incoming)

Redondo Beach, CA

- Incoming intern supporting software development for mission-critical defense systems.
- Maintaining active *Secret Security Clearance*.

Software Engineer Intern

polylabs.ai

June 2025 – Present

Palo Alto, CA

- Implemented social features in React and FastAPI for their educational user generated content platform which connected over 20,000 students
- Designed and created a statistics management system in Python for their UGC platform to track analytics such as views, user interactions, and engagement metrics
- Built a drag-and-drop AI-powered component builder for websites using React and Next.js, enabling dynamic UI generation and real-time preview

Founder & Lead Video Producer

JohansenVPN

Aug. 2022 – Jul 2024

Modesto, CA

- Created a photobooth application using C++ that interfaced real time video streams into a full stack photobooth application with touch screen support to ease user experience.
- Integrated *OpenCV* and *PyTorch* into the production workflow to automatically classify videos, using an algorithm that determines validity based on previous inputs.
- Secured \$100K from U.S Department of Education through leading my production team to create professional videos and engaging within the community.

Junior Software Developer

CraftiGames

Aug 2021 – Feb 2022

Remote

- Worked with a team of developers to create, design, and ship player-based games, taking the lead role in producing games.
- Increased revenue by 50% through player sales by implementing game designs that maximized revenue.
- Designed and deployed infrastructure changes that used modern technologies such as *gRPC* and *Redis*
- Improved efficiency within the backend infrastructure through a custom remote game system built with *Java*, *MySQL*, and *gRPC*, resulting in start times decreasing by 95%.

PROJECTS

tag2me.app | AI Reasoning, Video Processing at Scale

SaaS AI Media Analysis

- Built a web platform for analyzing social media content at scale (TikTok, Instagram Reels, YouTube Shorts) using AI to tag the content.
- Implemented a custom metadata extraction pipeline that extracted metadata such as (title, thumbnail, source) from social media platforms with inconsistent or restricted access.
- Designed a backend API using *Next.js* to support tagging, API access, and scalable content retrieval as well as integrated *Stripe* for subscription billing and payment management.
- Deployed monitoring and error tracking with *Sentry* to detect and resolve production issues.

Rice Stats | Docker, Grafana, Java, InfluxDB

Data Analytics Software

- Developed a plugin for Minecraft with *Java* that allows server owners to analyze their player interactions visually
- Deployed *Grafana* to visualize player data trends and behaviors, providing analytics to gain insight into players
- Leveraged *Docker* to scale *InfluxDB* efficiently to process thousands of data requests.

Frostless Network | Spigot API, Java, Git, UNIX, Postgres, DevOps

Game Infrastructure

- Designed, built, and deployed a custom multi-threaded Minecraft server which supported thousands of players with minimal latency.
- Focused heavily on optimization for both client and developer experience which involved using *Redis* for high availability caching and *PostgreSQL* for data persistence, reducing server load by 75%.
- Automated deployments and scaling with *Github Actions* and *TravisCI* to ensure real time updates with no downtime for clients.
- Engineered a game server system which handled networking, matchmaking and player data with the use of *Kubernetes*.
- Deployed and self managed bare metal servers using *TailScale*, improving developer productivity and ease of access during on call.

SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, HTML/CSS, Rust

Frameworks: React, Node.js, Tailwind, Next.js

Developer Tools: Git, Docker, Redis, TravisCI, Google Cloud Platform, RabbitMQ, Typescript

Awards: 1st place in Cyber Security Hackathon (2022); Modesto City Schools CTE Award